

Exoplanet Hunter — transit vetting with a calibrated dual-view CNN

Served model `ca906040`, promoted 2026-07-19

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1 Summary

This project vets transiting-planet candidates. It does not search for them: the input is a signal someone has already flagged — a TESS Object of Interest, a Kepler Object of Interest, a K2 candidate — with a published ephemeris, and the output is a calibrated probability that the signal is a planet rather than an eclipsing binary, a blend, or an instrumental artefact.

The served model is a dual-view 1-D CNN after Shallue & Vanderburg, trained with 5-fold cross-validation grouped by host star and calibrated per fold with Platt scaling. Out of fold it reaches ROC-AUC **0.9581 ± 0.0057** and Brier **0.0791 ± 0.0066** across five folds. On the mission that decides promotion, TESS, it recovers **31.1%** of real planets at a 1% false-positive rate; on Kepler, **81.3%**.

The larger part of the work was not the model that shipped but the eight architectural candidates that did not. An eleven-branch ExoMiner-inspired network was built, run, ablated, ensembled, extended twice, and closed. Every one of those readings was pre-registered before its numbers existed and read against a noise floor measured in the same run. None of them beat the model above on the metric that governs, so none of them was promoted, and the record of why is the substance of §4.

The single most important methodological commitment here is that **a margin smaller than its own noise floor is not a result**. Applied consistently, that rule closed a line of work this project had spent most of its compute on.

2 1. Problem formalisation

2.1 1.1 The task

Let a *candidate* be a periodic dimming signal in a star’s light curve, given as an ephemeris — period P , epoch t_0 , duration d — attached to a host star. Vetting is the binary decision

$$y \in \{\text{planet}, \text{not a planet}\}$$

made from the star’s photometry and its catalogue parameters. The model estimates $\Pr(y = \text{planet} \mid \mathbf{x})$ where

$$\mathbf{x} = (\underbrace{\mathbf{g} \in \mathbb{R}^{2001}}_{\text{global view}}, \underbrace{\boldsymbol{\ell} \in \mathbb{R}^{201}}_{\text{local view}}, \underbrace{\mathbf{a} \in \mathbb{R}^9}_{\text{auxiliary}})$$

These are the served model’s actual input signatures: `global_view [None, 2001, 1]`, `local_view [None, 201, 1]`, `aux [None, 9]`.

Framing it as vetting rather than detection is a real constraint, not a simplification. The ephemeris is given, so the model never has to find the period; equally, the model inherits whatever selection produced the candidate list, which is the source of the principal limitation in §7.2.

2.2 1.2 Cost structure, and what follows from it

The two errors are not symmetric and neither is cheap.

A **false positive** consumes follow-up: radial-velocity time on an oversubscribed spectrograph, or a night of ground-based photometry. A **false negative** loses a planet — not permanently, but it drops off the shortlist that anyone actually observes.

The scarce resource is follow-up capacity, and capacity is a *fixed budget*, not a threshold on probability. An observer does not ask “is this above 0.5”; they ask “I can afford 1% of my negatives as wasted nights — how many real planets does that buy me?” That question is exactly **recall at a fixed false-positive rate**, and it is the metric this project promotes on.

Two consequences run through the whole report:

1. **Recall @1% FPR is the decision metric.** ROC-AUC is reported beside it because it is far the more stable of the two — a bootstrap standard deviation of 0.0059 against 0.0410 on the same rows (§6.4) — and because it is the number this field prints. It is **not** reported because it is comparable to published work: an audit on 2026-09-10 found every candidate AUC comparison fails on population or protocol (§7.2), and an earlier draft of this line claimed otherwise. A model can also gain AUC while losing shortlist recall — stage 4’s capacity arm did exactly that (§4, row 4) — and when the two disagree, recall governs.
2. **Calibration is a first-class metric, not a nicety.** A shortlist is a *ranking under a budget*. If scores are not comparable across the list, the cut is arbitrary. Expected calibration error (ECE) and the Brier score are therefore reported beside every accuracy number, and the promotion gate can reject a model for degrading either.

2.3 1.3 Why accuracy is not reported as a headline

Two independent reasons.

The labelled set is near-balanced **by construction** — 2,985 positive against 2,827 negative (§2.2) — because certain negatives are as hard to obtain as confirmed planets. That balance is an artefact of labelling, not the deployment prior.

And at deployment the prior is extreme in the other direction. Of the 3,919 TESS candidates the pipeline could actually process and score, **82.85%** score at or above 0.5. A classifier that answered “planet” to everything would post an accuracy in the eighties on that population while being useless for the decision it exists to support. Accuracy measures the wrong thing here; the shortlist metrics in §6 measure the right one.

3 2. Data collection

3.1 2.1 Sources

Source	What it provides
NASA Exoplanet Archive — TOI, KOI (DR25), K2 tables	dispositions, ephemerides, stellar parameters
ExoFOP	community TOI/CTOI dispositions and comments
MAST, via <code>lightkurve</code>	SPOC and Kepler stitched light curves
Gaia DR3	crossmatch for stellar parameters

Provenance, per-table row counts and the sky coverage of each are in [data_provenance.md](#). Raw FITS are treated as an evictable cache of immutable archive files; everything derived is rebuilt by current code and versioned in DVC (§8.2).

3.2 2.2 The labelled set

`data/tables/labels/labels.parquet` — 5,812 rows, one per host TIC, no duplicates.

Mission	Negative	Positive	Total
TESS	1,362	1,420	2,782
Kepler	1,250	1,250	2,500
K2	215	315	530
All	2,827	2,985	5,812

The Kepler counts are a configured cap, not the available population. The exact 1,250 / 1,250 is the signature: `catalog.py::_stable_sample` ranks rows by `md5(seed:tic_id)` and takes the first n , so the subsample is unbiased with respect to anything astrophysical and stable across refreshes — but it is a subsample. Roughly 2,748 eligible confirmed Kepler planets and 3,813 eligible false positives exist under the same criteria, so the served model was trained on about 45% less

Kepler data than was available to it. That is a deliberate balance choice, and it costs statistical power precisely where §6.4 shows this project has none to spare. Any comparison of this project’s set size to a published one should quote both numbers.

Label rules:

Disposition	Label
CP, KP (TESS); CONFIRMED (Kepler, K2)	1
FP, FA (TESS); FALSE POSITIVE, REFUTED (Kepler, K2)	0
PC, APC, CANDIDATE	held out — never trained on, never evaluated on

Kepler negatives are restricted to DR25 FALSE POSITIVE KOIs with a Robovetter disposition score below 0.5 — a majority false-positive vote across the Robovetter’s perturbed runs [thompson2018]. **This is not the Kepler Certified False Positive table** [bryson2015], which rested on Follow-up Observation Program working-group vetting against follow-up data; that table is no longer served by the archive over TAP or its retired legacy API, so `data/catalog.py::_query_certified_fp` reconstructs the criterion from the DR25 KOI table instead. About 79% of the bare cumulative false positives pass it — 3,813 of 4,839. The 1,026 that do not are dropped almost entirely on DR25 membership rather than on the score: 893 are absent from the DR25 table and 129 are dispositioned CANDIDATE there, against **four** that are DR25 false positives failing the 0.5 threshold. The threshold is close to inert, because `koi_score` is near-degenerate on the class it filters — 89% of DR25 false positives score below 0.001. Measured 2026-09-10; see the [comparability audit](#) §2.

So a negative here means *the Robovetter rejected it consistently under perturbation* — stronger than “not yet confirmed”, weaker than “certified by the FPWG against follow-up observations”. That distinction matters because without any such restriction the negative class would be contaminated with undiscovered planets and every recall figure in this report would be optimistic by an unknown amount; with the reconstruction it is contaminated only to the extent the Robovetter is wrong. See [data_provenance.md](#).

The served model saw no K2 rows. Of the 5,812 labelled rows, 4,818 carry an out-of-fold prediction from the served run; joining those predictions to the current catalogue gives TESS 2,371 and Kepler 2,237, **K2 zero**, with 210 rows whose `tic_id` no longer joins a label row because the catalogue has been refreshed since the run was trained. The 530 K2 rows were added to the labelled set after `ca906040` was trained and have never been part of a promoted run’s population. Every K2 count in this section describes the data on disk today, not the data the served model was fitted on, and no result in §6 is a K2 result.

3.3 2.3 The held-out candidate catalogue

`data/tables/catalogue/candidates.parquet` — 12,472 rows, TOI 8,148 and CTOI 4,324, spanning dispositions PC 4,828, FP 1,303, CP 816, KP 607, APC 485, FA 102. This is the population the live console lists and the deployed model scores. It overlaps the labelled set on the confirmed and false-positive rows; the 5,104 PC/APC rows that are not in the labelled set are the genuinely unresolved candidates the project exists to rank.

4 3. Preprocessing

The full specification is `model_specs.md` §1; configuration is `pipeline/conf/preprocess/default.yaml`. The steps, and why each is there:

1. **Stitch and normalise.** SPOC/Kepler PDCSAP flux, stitched across sectors or quarters, normalised per segment.
2. **Detrend with the transit masked.** A Savitzky-Golay filter removes stellar variability. The in-transit points are excluded from the fit, so the detrending cannot absorb the dip it exists to preserve — a filter fitted through the transit will happily flatten it.
3. **Phase-fold** at the catalogue ephemeris where one is published, from a BLS search otherwise.
4. **Median-bin** to two views: `global_view`, 2001 bins over the full phase $[-0.5, 0.5]$, and `local_view`, 201 bins over ± 3 transit durations about phase zero. The global view carries the shape of the orbit and any secondary eclipse; the local view carries the transit’s own profile at resolution the global view cannot afford.
5. **Auxiliary vector**, built by `features/aux.py::build_aux_row` — the single implementation shared by training and serving, so the two cannot drift. Imputed, log-transformed on the heavy-tailed columns, then standardised, with the transform fitted per fold and persisted in that fold’s calibration bundle.

The served model consumes the 9-dimensional layout: `teff`, `radius`, `logg`, `tmag`, `depth`, `duration`, `log_period`, `pink_snr`, `centroid_snr`. The current build extends this to 13 with four scalar vetting diagnostics (`oe_depth_sigma`, `oe_timing_sigma`, `secondary_sig`, `q_ratio`); `LEGACY_AUX_DIM = 9` keeps older runs loadable bit-identically.

A measured null worth keeping. Adding those four features changed ROC-AUC by -1.3×10^{-5} — indistinguishable from zero. The information was already present in the folded views, and the features were *scalar summaries* of diagnostics whose signal lives in shapes. That null is the origin of the branch architecture in §4: feed the odd/even, secondary and centroid **views** rather than statistics computed from them. It is also the first case in the project of a plausible improvement being measured and found empty.

5 4. Model selection

5.1 4.1 How a candidate was allowed to win

Every architecture below was judged by the same procedure, in this order:

1. **Pre-register the reading.** What statistic, on what rows, against what comparator, and what result would falsify the hypothesis — written down and committed before the run produced numbers. The pre-registrations are files in `experiments/`, each sitting immediately before the result it governs.
2. **Measure the noise floor in the same run.** With n members trained per fold, the floor is $2\sigma/\sqrt{n}$ over members. A margin inside its floor is not a result.
3. **Run the promotion gate.** `promotion_gate.py` compares to the champion on out-of-fold TESS ROC-AUC, with Brier not degrading by more than 0.005 and ECE by more than 0.01, and recall @1% FPR not falling by more than the run’s own measured floor. It returns one

of three verdicts: **PROMOTE** (exit 0), **REJECT** (1), **UNRESOLVED** (2) — the last meaning the margin is inside the floor and the question is unanswered, which is a stop-and-ask rather than a rejection.

4. **Read the result exactly as pre-registered.** A result landing outside the terms fixed before it ran is recorded as falsified and never re-specified.

The gate has rejected several retrains. That is it working.

5.2 4.2 The record

Every row traces to a frozen file in [experiments/](#). “Margin” is against the comparator in the same row; “floor” is the noise floor measured in that run, by the rule above.

#	Candidate	Decision metric	Measured	Comparator	Floor	Verdict
1	Random forest on 14 hand-crafted features	—	—	—	—	never scored
2	Dual-view CNN (served)	TESS recall @1% FPR	0.3069	—	—	champion
3	Branch model, runs 1–3	TESS recall @1% FPR	0.238 / 0.126 / 0.145	0.307	0.034	rejected
4	Branch + capacity	TESS recall @1% FPR	0.236	0.307	0.034	rejected
5	Leave-one-family-out attribution sweep	per-arm Δ AUC	null on every arm	—	—	no arm promoted
6	Control-arm host test (stage 7i)	host-AUC advantage	none measured	incumbent	—	criterion not met
7	Propensity weighting (stage 8)	baseline-dependence gap	−0.1336	0	3.3× bar	bias fix, confirmed
8	Dual-view + branch ensemble (10.5)	shortlist recall	0.4362 / 0.4223	0.3046	—	complement, not promoted

#	Candidate	Decision metric	Measured	Comparator	Floor	Verdict
9	Difference- image branch (stage 9)	control- arm host-AUC	unmeasurable	—	—	no effect shown
10	Phase 1 — target- position channel	TESS recall @1% FPR, dv_usable	−0.0254	arm C''	0.0979	falsified (0.26×)

5.2.1 Row 1 — the baseline this project does not have

`models/baseline_rf.py` and `conf/model/random_forest.yaml` are in the repository, with a documented rationale, and the classical baseline is described in the predecessor project’s report. **No scored cross-validated result for it exists anywhere in this repository:** `mlflow.db` holds 197 runs and every one that records a model name records `cnn_dualview`. It is reported here as absent rather than quietly omitted, and it is listed in [known-limits.md](#) as an outstanding gap. The honest reading is that the CNN’s advantage over classical ML is *assumed* in this project, not measured.

5.2.2 Row 2 — the model that shipped

A dual-view 1-D CNN after Shallue & Vanderburg [shallue2018]: two convolutional towers, one per view, concatenated with the auxiliary vector and passed through a dense head, with Squeeze-and-Excitation channel attention [hu2018] placed after each conv block and before pooling, multi-head attention pooling, and a residual late-fusion head. Full evaluation in §6.

Platt scaling [platt1999] rather than temperature scaling [guo2017], for a measured reason: temperature is a single parameter applied to the logit and has no bias term, so it cannot shift the decision boundary under a distribution shift. Platt’s affine form can. Both are rank-preserving, so neither changes AUC — the difference is entirely in calibration, and using the wrong one cost a 0.136 ECE regression before it was diagnosed.

5.2.3 Rows 3–4 — the branch model, and why more capacity did not save it

The branch model is an ExoMiner-inspired rebuild: eleven views, one convolutional branch each, reduced to per-branch embeddings and fused in a shared head. Its structural argument over the dual-view design is **unfolded_view** — up to 20 individual transits fed unstacked, so the network can see whether a signal *recurs* rather than only the average of everything that recurred.

Across three runs it scored 0.238, 0.126 and 0.145 TESS recall @1% FPR against the champion’s 0.307, on a decision floor of 0.034 — rejections at up to 4.8× the floor, not close calls. Its resolution hypothesis was falsified separately (Kepler moved +0.0707, the wrong direction for the proposed mechanism), and the capacity arm closed the “it is simply too small” explanation: paired difference −0.0035, $d = -0.44$. That arm is also the clearest case of AUC and shortlist recall disagreeing — recall rose 0.145 → 0.236 while AUC fell.

5.2.4 Row 7 — the one arm that changed something, and did not promote

Stage 8 tested propensity weighting against the observation-baseline confound (§7.2). It worked: the branch model’s amplification of baseline dependence was removed, a gap of -0.1336 at $3.3\times$ its bar, at no cost in recall. It is recorded as a **bias fix**, not a promotion, because it does not improve the decision metric — it removes a reason to distrust the model’s ranking. All four of the stage’s own predictions were falsified; the win it did produce was not one it had predicted.

5.2.5 Row 8 — the complement finding

Ensembling the dual-view and branch models reached shortlist recall 0.4362 and 0.4223 against the common-fold dual-view member’s 0.3046. That is a real gain, and it is the strongest argument in the record that the branch line was learning *something* the dual-view model was not.

It did not promote, for a reason worth stating plainly: an ensemble of two architectures is not a candidate for the champion slot under this project’s serving constraints — it doubles inference cost on a scale-to-zero deployment, and the gate compares single models to the champion. It is recorded as a complement finding: evidence that the branch’s information is not redundant, banked for a future design rather than shipped.

5.2.6 Rows 9–10 — closing the line

Stage 9 added a difference-image branch — pixel-level source location, the strongest single nearby-eclipsing-binary discriminant. Its falsification test was **unmeasurable** as specified, because the control-arm harness zeroes the branch on exactly the rows the test needed. Three secondary predictions confirmed it cost nothing; none showed it delivered anything.

Stage 9’s readout named one remaining explanation for that null: the stamps were fed without the star’s own pixel position, so the network had no reference frame in which a centroid shift means anything. Phase 1 built that channel and ran the paired arms C’’ (branch dropped at the model) and D’’ (branch kept), on identical augmented batches.

On the pre-registered statistic — TESS recall @1% FPR on `dv_usable` rows, $n = 2,077, 1,220$ positive:

	arm C’’	arm D’’	D – C	floor	ratio	verdict
recall @1% FPR	0.2197	0.1943	−0.0254	0.0979	0.26×	within floor
ROC- AUC	0.9228	0.9188	−0.0040	0.0154	0.26×	within floor

Two floors could have governed — a member-pairing floor and the Phase 1a seed floor of 0.0432 — and which one would be used was fixed in writing at 03:52 while arm D’’ was still training, with an explicit rule that disagreement between them would be recorded UNRESOLVED. They agreed; the margin is inside both.

Read exactly as pre-registered, the reference-frame explanation is falsified. No third stamp variant was commissioned and the branch line is closed in [decisions.md](#). Two things are

stated because omitting either would flatter the result: the margin is negative in every cell, and it is not large enough to be called harm either — the honest reading is *no effect*. And one post-hoc number is recorded and explicitly not banked: TESS ECE improved $0.0342 \rightarrow 0.0238$ from C'' to D'', with no ECE floor pre-registered.

5.3 4.3 What the record shows

Ten candidates; one served; one bias fix adopted; one complement finding banked but not shipped; one line of work closed by its own pre-registered criterion. The compute spent on rows 3–10 bought no improvement to the served model. It bought something else: a documented reason to believe the served model is not being beaten by an obvious alternative, and a demonstration that this project's gate cannot be talked past.

6 5. Training

6.1 5.1 Cross-validation

5-fold StratifiedGroupKFold, grouped by host star, stratified on label.

The grouping is inert on this labelled set, and an earlier draft of this report credited it with work it does not do. The splitter is passed `tic_id` as the group key, but `labels.parquet` holds 5,812 rows on 5,812 distinct `tic_id` — every group has size one, so `StratifiedGroupKFold` produces the same partition `StratifiedKFold` would. Verified 2026-09-10; the maximum group size is 1 and no host appears twice.

This project is nonetheless free of the multi-planet leakage the grouping exists to prevent, but the mechanism is upstream: the label builder emits one row per host, so a star cannot be split across folds because a star cannot appear twice at all. The splitter is a correct guard that currently has nothing to guard — worth keeping, since a future TCE-level labelled set (deferred item #37) would make it load-bearing overnight, but it earns no credit for the current numbers.

The cost of the dedup is real and is not stated elsewhere: collapsing to one row per host discards the multi-planet systems' additional signals. It is a correctness choice with a power price, not a free one.

Within each outer fold an inner 88/12 `GroupShuffleSplit` separates training from validation. The inner validation set drives three things and is never used for reported metrics: early stopping, the F1-optimal decision-threshold sweep, and the calibrator fit.

6.2 5.2 The served run

Champion `ca906040cdb74ba6b07353a500244777`, promoted 2026-07-19, unchanged since. Its five folds are MLflow child runs of that parent:

Fold		Epochs recorded
fold-0	none — no per-epoch series was logged	
fold-1		78
fold-2		79
fold-3		70

Fold	Epochs recorded
fold-4	84

Early stopping on inner-validation loss accounts for the spread. Fold-0’s missing series is a logging gap, not a training failure: its weights, its out-of-fold predictions and its calibration bundle are all present and it contributes to every pooled number in §6. The training-curve figure names the missing fold in its title rather than silently drawing four lines and calling them five.

6.3 5.3 Calibration and uncertainty

Platt scaling is fitted per fold on the inner validation logits: the fold’s mean intercept is $b = 0.942 \pm 0.554$ and slope $a = 0.916 \pm 0.087$. The fitted decision threshold is 0.486 ± 0.039 — near 0.5, which is what near-balanced labelled data should give, and reported rather than assumed.

At serving, uncertainty comes from Monte-Carlo dropout: repeated stochastic forward passes give `prob_mean`, `prob_std` and a p10–p90 interval per candidate, all surfaced on the console. Fold disagreement is reported separately from MC spread, because they measure different things — the console labels them `Total sigma (MC + fold)` and `Within-fold sigma` rather than showing one number twice.

6.4 5.4 Ensembling, and the floor

Both trainers accept `n_models_per_fold`. Members are trained independently, their scores averaged, and the calibrator fitted once over the average; each member’s uncalibrated score is written out, so the spread the run averaged over stays readable afterwards. **That spread is the noise floor every margin in this project is read against.**

The served model trains **one member per fold**, so it has no seed spread of its own and therefore **no measured noise floor**. The API reports this in exactly those words rather than substituting a floor measured on a different architecture — a branch-model floor printed under dual-view numbers would be a category error, and fixing an instance of it was a deliberate step of this work.

7 6. Evaluation

7.1 6.1 Protocol

Every number in this section is **out of fold**: each row is scored by the fold that did not train on it. There is no additional held-out test set, and this is a deliberate trade — with 5,812 labelled rows, a 15% holdout would be ~870 rows, too few to resolve a per-mission recall difference at the floors this project reads against. The cost is that no number here is protected from the model selection in §4, and the mitigation is the pre-registration discipline: readings were fixed before results existed.

Pooled predictions: 4,818 rows, 2,551 positive.

7.2 6.2 Cross-validated performance

Fold means $\pm$ standard deviation across the five folds, from `models/cv/ca906040.../cv_summary.json`:

Metric	Mean	SD
ROC-AUC	0.9581	0.0057
PR-AUC	0.9599	0.0048
F1 (at the fitted threshold)	0.9001	0.0125
Brier	0.0791	0.0066
ECE	0.0276	0.0048
Decision threshold	0.486	0.0388

7.3 6.3 Per mission, at the operating point

TESS decides promotion; Kepler is diagnostic. K2 does not appear because the served run contains no K2 rows (§2.2). Each mission’s confusion matrix is cut at **its own 1% FPR threshold**, and the recall printed beside it is the recall at that same cut — one measurement, not two.

	TESS (gating)	Kepler (diagnostic)
n / positive	2,371 / 1,304	2,237 / 1,245
ROC-AUC	0.9100 $\pm$ 0.0088	0.9914 $\pm$ 0.0028
Recall @1% FPR	0.3113 $\pm$ 0.0656	0.8129 $\pm$ 0.0511
Brier	0.1210 $\pm$ 0.0092	0.0360 $\pm$ 0.0055
ECE	0.0435 $\pm$ 0.0134	0.0407 $\pm$ 0.0039
Threshold at 1% FPR	0.9612	0.8314
Actual FPR at that cut	0.0103	0.0101
TP / FP / FN / TN	406 / 11 / 898 / 1,056	1,012 / 10 / 233 / 982
Precision at that cut	0.9736 $\pm$ 0.0066	0.9902 $\pm$ 0.0025
F1 at that cut	0.4718 $\pm$ 0.0734	0.8928 $\pm$ 0.0306

Recall @5% and @10% FPR, on the gate’s re-scored population (see the note below): TESS 0.5608 and 0.7308; Kepler 0.9575 and 0.9888.

Two populations, one run. The record and the promotion gate quote TESS recall @1% FPR as **0.3069** on 2,367 rows, from `models/cv/champion-rebaselined-today/cv_summary.json` — the champion re-scored out of fold on the labelled set as it stood on 2026-08-17. The API computes **0.3113** on 2,371 rows, joining the same predictions to the labelled catalogue as it stands today. Same model, same predictions, four rows of difference in the join. Both are reported here; §4’s table uses the gate’s number because that is the number promotion decisions were made against.

7.4 6.4 Reading those numbers

The Kepler–TESS gap is the headline finding of the evaluation. ROC-AUC 0.991 against 0.910, and recall @1% FPR 0.81 against 0.31, on the same model at the same time. Kepler stared at one field for four years; TESS mostly gets 27 days per sector. The transits are shallower, the baselines shorter, the systematics worse. A model reported only on its pooled AUC of 0.958 would

look far more capable than it is on the mission that actually matters for new discoveries. This is why TESS is the gating mission and why per-mission slicing is not optional.

Precision is high and F1 is low on TESS, and that is expected. At a 1% FPR cut the model makes 417 positive calls of which 406 are right — but there are 1,304 real planets, so it misses 898. F1 balances precision and recall as though they were equally valuable; under a fixed follow-up budget they are not. F1 is reported for completeness and is not a promotion criterion.

The decision metric is an eight-to-ten row statistic, and that governs how every margin in §4 should be read. At a 1% false-positive rate on 1,067 TESS negatives the threshold is the 10th-highest negative score; on the Phase 1 `dv_usable` population it is the 8th. A paired bootstrap of the Phase 1 contrast gives a sampling sd of **0.0437** for the difference, so twice that — 0.087 — is the same order as the seed floors the record reads margins against. Three members of a single arm, identical data and architecture, span 0.1418 to 0.2221.

The consequence is stated plainly: **any architecture effect smaller than about 0.09 in TESS recall @1% FPR is undetectable at this sample size, however many models are trained.** The rejections in §4 rows 3–4 are real, at up to 4.8x their floor. The nulls in rows 5–10 are mostly *underpowered*, not *negative*, and the promotion gate returning UNRESOLVED rather than REJECT is the gate correctly saying so. ROC-AUC, by contrast, has a bootstrap sd of 0.0059 on the same rows — seven times more stable — because it uses every row rather than one crossing point. Any future comparison on this project should be powered before it is run.

Calibration is good pooled and mediocre per mission. Pooled ECE is 0.0121; per mission it is 0.0435 (TESS) and 0.0407 (Kepler). Pooling cancels errors of opposite sign — TESS is over-confident where Kepler is under-confident — so the pooled figure flatters. The per-mission figures are the ones to read.

7.5 6.5 Figures

All regenerated from the served run by `pipeline/scripts/make_performance_figures.py`.

Figure	What it shows
<code>figures/roc_operating_point.png</code>	Per-mission ROC at unit aspect, 1% FPR operating point marked
<code>figures/roc_pr.png</code>	Pooled ROC and precision–recall curves
<code>figures/calibration.png</code>	Reliability diagram, 10 bins, with the diagonal
<code>figures/training_curves.png</code>	Per-fold loss and AUC by epoch; the title names the fold with no series
<code>figures/embedding_3d.png</code>	Penultimate-layer embedding, coloured by label
<code>figures/risk_coverage.png</code>	Risk against coverage as the score threshold sweeps
<code>figures/completeness.png</code>	Recovery against transit depth and period
<code>figures/sky_map.png</code> , <code>figures/coverage_map.png</code>	Sky and sector coverage of the labelled set

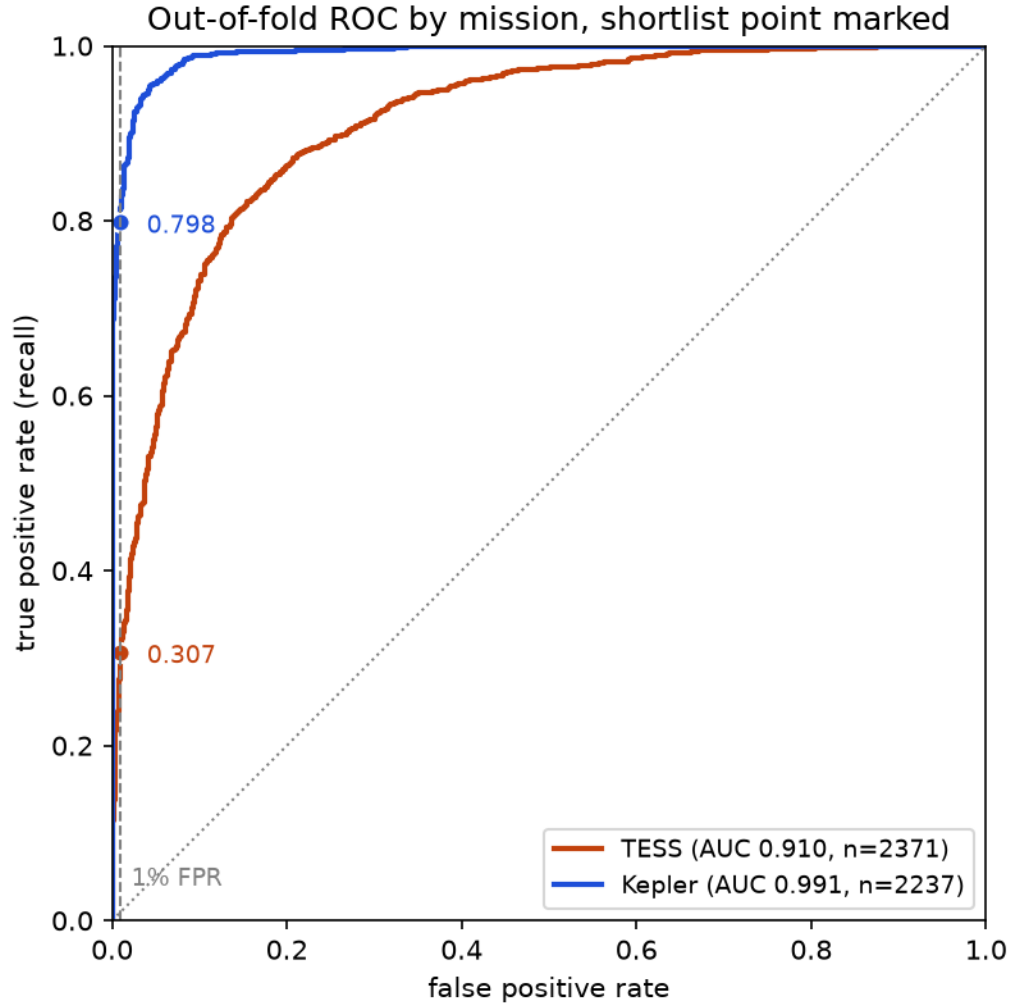


Figure 1: Out-of-fold ROC by mission, drawn at unit aspect with the 1% FPR shortlist point marked. The marked values, 0.307 on TESS and 0.798 on Kepler, use the conservative cut — the highest recall reachable *without* exceeding the budget — which is the convention the promotion gate uses (§6.3).

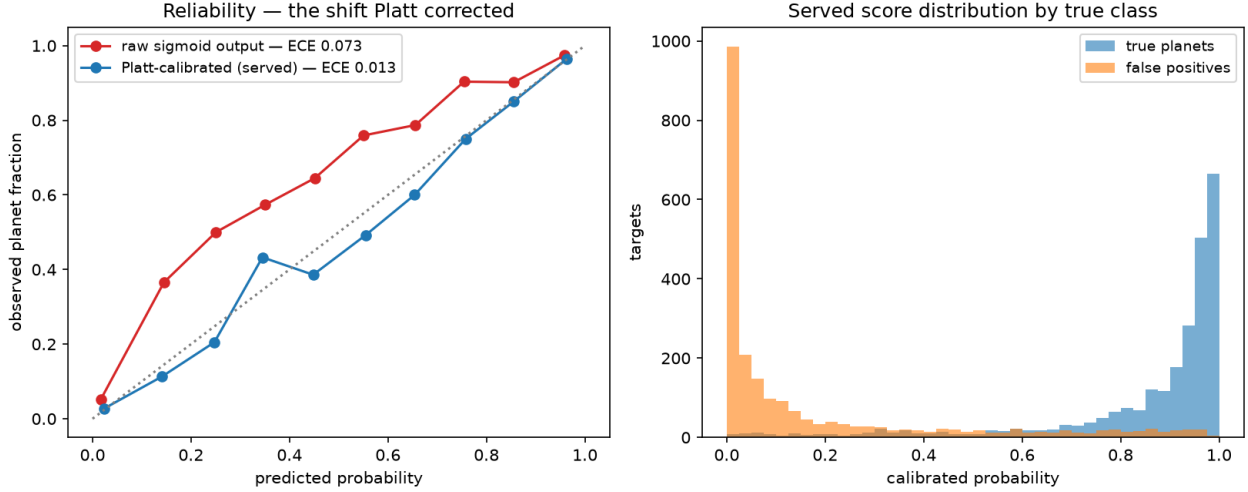


Figure 2: Reliability diagram: predicted probability against observed frequency, ten bins, pooled out of fold.

8 7. Results and limitations

8.1 7.1 What the deployed system produces

The API scores the held-out candidate catalogue in bulk and serves the results to the console. On the most recent bulk pass, of 4,685 candidates attempted:

Outcome	Count
Scored	3,919
No light curve available at MAST	744
Preprocessing failed	22

Of the 3,919 scored, **82.85%** score at or above 0.5 and **1.28%** at or above 0.9. That distribution is itself a result: the candidate list is dominated by signals that already look planet-like, which is why a 0.5 cut is nearly vacuous on this population and why the console’s default ranking is by score under an explicit budget rather than by a fixed threshold.

8.2 7.2 Limitations

The observation-baseline confound (W1). How long a star was watched correlates with its label: on TESS the correlation between observation baseline and label is +0.387. Longer baselines produce more confirmations, so a model can gain apparent skill by learning observing strategy rather than astrophysics. It is measured, not hypothesised: the branch model amplified the correlation to +0.5155, which is 0.13 *above* the labels’ own. Stage 8 showed propensity weighting removes the amplification (§4, row 7). The confound in the labels themselves remains, and every recall figure in this report is read under it. The console now shows the derived baseline on each candidate row so a reader can see the confound rather than take it on trust.

The model scores the star, not only the signal (W2). Some of the model’s discrimination comes from host properties rather than from the transit. The control-arm harness that would quantify this zeroes the branch on exactly the rows that would test it, which is why stage 9’s primary criterion was unmeasurable. A control-arm host-AUC floor was measured for the first time in Phase 1a — 0.0198, from three seeds at mean 0.5826 — so the instrument now exists even though the stage-9 question stayed unanswered.

No classical baseline. §4, row 1. The CNN’s advantage over hand-crafted features is assumed here, not measured.

No measured noise floor for the served model. One member per fold (§5.4). Every floor quoted in this report was measured on a different run, and is used only to read that run’s own margins.

The unexplained Kepler cell (W7). A +0.1446 movement in one Kepler cell has never been explained. It is recorded rather than resolved.

Unbounded cross-validation runtime (W10). Fold duration within a single Phase 1 arm ranged from 25 to 78 minutes with no bound on the slowest fold, which makes compute budgeting unreliable for any future sweep.

Evaluation is out of fold, not on a fresh holdout. §6.1.

The Kepler set is a 45% subsample of what was available, and the host-level dedup that makes the split leak-free also discards every multi-planet system’s additional rows. §2.2 and §5.1. Both cost power.

This project is behind the published state of the art on the metric it promotes on, and the size of the gap is now bounded. A benchmark-comparability audit on 2026-09-10, whose six verdicts were then put through a three-refuter adversarial panel, found that of six candidate comparisons to published work, five fail on population, protocol or metric definition and cannot be stated in either direction.

The sixth can. An earlier draft of this section claimed that recall @1% FPR — the metric promotion is decided on — has no published counterpart. **That was wrong, and the correction is unfavourable.** ROC curves are non-decreasing, so any published operating point at an FPR below 1% is a valid lower bound on recall @1% FPR, and two of the five sources publish one:

source	published point	implied FPR	bound	ours
ExoMiner	recall 0.936 at	0.077%	0.936	0.8129
2022 (Kepler)	99% precision			
ExoMiner++ (TESS)	recall 0.951	0.47%	0.951	0.3113

This project’s own realised cuts are stricter still — 0.94% FPR on TESS, 0.70% on Kepler — so the inequality runs the right way and the comparison is valid one-sided: roughly **12 points behind on Kepler and 64 points behind on TESS**. Negative-class composition affects the magnitude, not the sign.

It is stated here because the alternative is not rigour. The audit’s original refusal to compare read as methodological care; on the one metric that governs promotion it was suppressing a 64-point deficit, and the refuting agent said so. See [experiments/benchmark-comparability-2026-09-10.md](#) §8.

The decision metric has too little power to settle architecture questions. §6.4. This is the limitation that most shapes what §4 can and cannot claim.

The full register, with each item’s evidence and status, is [known-limits.md](#).

9 8. Reproducibility

9.1 8.1 Environment and commands

```
make env      # conda env from environment.yml, editable installs
make test     # fast suite (network and slow markers excluded)
make lint     # ruff
make type     # mypy against the tracked baseline
make validate # data validation gates on whatever artefacts exist
make refresh  # the full refresh DAG; trains only if warranted
make report   # render docs/report.md to docs/report.pdf
```

9.2 8.2 Data and model versioning

Artefacts are DVC-tracked with a Cloudflare R2 remote: `.dvc` pointer files are in git, the bytes are in R2. `make data-pull` materialises them; `make data-push` syncs. The promoted model is pinned in `models/registry.json`, which carries the run id, its `cv_summary.json` path and the metrics it was promoted on. That file is not edited by hand — only the promotion gate writes it.

9.3 8.3 Experiment tracking

MLflow (`mlflow.db`, `mlruns/`), one parent run per cross-validation with a child run per fold. Every number in §6 comes from either `cv_summary.json` or `predictions.parquet` under `models/cv/<run_id>/`.

9.4 8.4 Serving

FastAPI on Fly.io, scale-to-zero, with the console a single static build on Render that deploys from `main`. The API deploys manually:

```
fly deploy --remote-only
```

The `/score/{tic_id}` contract is pinned: `api/app/schemas.py` and the console’s client `frontend/design-console/src/app.api.js` change together or not at all, and a contract test extracts every field the client reads and asserts it exists on the Pydantic model, so the two cannot drift silently.

9.5 8.5 The record

[experiments/](#) holds one frozen file per stage or audit, in the order things happened, with each pre-registration immediately before the result it governs. Nothing there is rewritten: a correction is a dated note appended under the entry it corrects. Every number in §4 traces to one of those files.

10 9. References

Full bibliography in [references.bib](#).

Key	Why it is cited here
shallue2018	the dual-view architecture the served model follows
valizadegan2022, valizadegan2025	ExoMiner and ExoMiner++ — the per-diagnostic branch design §4 is <i>inspired by</i> , not a reimplementation of
thompson2018	Kepler DR25 and the Robovetter disposition score that defines the negative class (§2.2)
bryson2015	the Kepler Certified False Positive table — cited to say what this project’s negative class is not
twicken2018	the DR25 transiting-planet search that produced the KOI population
hu2018	Squeeze-and-Excitation channel attention, in the served architecture
guo2017, platt1999	temperature scaling and Platt scaling — why the calibrator is the affine one
jenkins2016	the SPOC pipeline that produces the light curves
giacalone2021	the vetting problem as the community frames it
lightkurve2018	the library the ingest layer uses
platt1999	Platt scaling, the calibrator fitted per fold
gal2016	MC dropout, the serving uncertainty estimate
nasa, exofop	the catalogues the labels come from